



BOILEAU GUILLAUME

Architecte IA / Senior Data Scientist | Python, ML, LLMs, MLOps foundations, systèmes complexes

📧 guillaume.boileaupro@gmail.com
📧 Guillaume-Boileau-2

📞 +33 6 59 94 85 84
📞 0000-0002-3576-6968

📍 Alpes-Maritimes, FRANCE

📄 guillaume-boileau-891b81265

EXPERIENCE

Software Engineer – AI-assisted development, integration & operational validation

VEV Platform Services France

📅 2025 – 2026

📍 Valbonne, France

Context: Industrial software environment for an EV charging CPMS platform connected to operational charging stations, backend services, field equipment and hardware/software interfaces.

Objective: Contribute to software delivery, integration and operational reliability by combining development, data-flow analysis, incident investigation, technical documentation and AI-assisted engineering practices.

- **Software delivery:** Contributed to TypeScript, Angular and Node.js components integrated into an operational software platform.
- **AI-assisted workflows:** Used AI tools for code review, debugging support, documentation, incremental problem solving and productivity improvement.
- **System integration:** Worked on interfaces between software services, operational data, connected infrastructure and field constraints.
- **Operational data flows:** Analysed FTP-based exchanges, interrupted flows, inconsistent data and service-impacting behaviours.
- **Validation:** Checked consistency between application data, expected behaviours and operational constraints.
- **Monitoring and support:** Investigated incidents, documented technical findings and contributed to service reliability improvements.
- **Agile coordination:** Used Zenhub for task tracking, backlog follow-up, prioritisation and coordination in an iterative environment.

Senior R&D Engineer – Scientific AI, model validation & Python platform development

CNES

📅 2023 – 2025

📍 Nice, France

Context: R&D and engineering activities for the LISA ESA/NASA space mission, involving simulation, scientific Python, statistical modelling, signal processing, model integration and validation workflows.

Objective: Design and validate reproducible Python workflows able to integrate heterogeneous models, compare outputs, assess weak signals under noise and produce traceable results for collaborative technical review.

- **Python platform:** Developed CATpyLISA, a Python platform for model integration, comparison, validation and technical review. [GitLab: CATpyLISA](#)
- **Data science workflows:** Structured datasets, simulations, model outputs, validation logic and reproducible analysis pipelines.
- **Statistical modelling:** Worked on low signal-to-noise physical backgrounds, uncertainty analysis and model comparison.
- **Machine learning mindset:** Applied data-driven reasoning, feature interpretation, validation metrics and reproducibility principles to complex scientific data.
- **Model validation:** Defined consistency checks, comparison rules, acceptance criteria and validation procedures for heterogeneous contributions.
- **Technical coordination:** Interfaced with scientific, software and institutional contributors in an international space-mission environment.
- **Documentation and knowledge sharing:** Used Confluence-style documentation logic for traceability, review preparation and collaborative knowledge transfer.
- **Decision support:** Produced technical syntheses, identified discrepancies and formalised recommendations for model and workflow improvements.

Data Scientist – Statistical modelling, noise analysis & ML-oriented workflows

Universiteit Antwerpen, Belgium

📅 2021 – 2023

📍 Antwerp, Belgium

Context: Data science and performance studies for next-generation precision instruments, combining environmental sensor data, statistical modelling, weak-signal detection, noise reduction and validation of complex systems.

Objective: Develop robust and reproducible workflows to identify, characterise and reduce noise sources affecting measurement quality and system performance.

- **Data science:** Developed Python workflows for signal analysis, statistical modelling, performance assessment and validation.
- **Bayesian modelling:** Built MCMC-based pipelines to analyse different detector configurations and noise scenarios.
- **Sensor data:** Analysed seismometer and environmental data, propagation effects, spatial correlations and noise impacts.
- **Machine learning:** Structured approaches for non-stationary noise reduction using witness channels, machine learning and deep learning.

PROFILE

PhD physicist and senior data scientist with experience in Python, statistical modelling, machine learning, signal processing, model validation and software workflows for complex systems. Strong background in scientific AI, reproducible pipelines, low-SNR data, technical coordination and knowledge transfer.

Positioning for this role: AI architect / senior data scientist able to structure AI use cases from ideation to validated POCs, support industrialisation with MLOps foundations, and help teams adopt LLM, predictive analytics and data-driven workflows.

- Strong quantitative background: statistics, modelling, signal processing, uncertainty analysis and validation.
- Python development for reproducible data-processing, simulation and model-validation pipelines.
- Experience with ML-oriented workflows, deep learning concepts, LLM/agent AI tools, chatbots and MCP-based architectures.
- Able to bridge technical architecture, data science, software delivery, documentation and stakeholder acculturation.
- Fast ramp-up profile for Azure ML, Databricks, MLflow and enterprise LLMops environments.

LANGUAGES

English



Spanish



EDUCATION

PhD in Physics

Université Côte d'Azur

📅 2018 – 2021

📍 Nice, France

Selective Graduate Program in Fundamental Physics (Magistère)

Université Paris-Saclay

📅 2016 – 2018

📍 Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

📅 2017 – 2018

📍 Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

📅 2015 – 2016

📍 Orsay, France

PROFESSIONAL TRAINING

Machine Learning in Python with scikit-learn

FUN MOOC – Inria

📅 2026

📍 France Université Numérique

Predictive modelling, supervised learning, model evaluation, cross-validation, metrics, pipelines and practical machine learning workflows with Python and scikit-learn.

- **Performance metrics:** Used figures of merit, sensitivity studies and validation checks to compare configurations and quantify robustness.
- **Technical reporting:** Produced traceable results, technical syntheses and recommendations for international stakeholders.

Junior R&D Engineer – Signal processing, simulation & scientific computing

ARTEMIS, Observatoire de la Côte d'Azur

📅 2018 – 2021

📍 Nice, France

Context: Scientific computing and signal-processing activities for gravitational-wave mission preparation, involving simulation, weak-signal separation, diagnostic analysis and uncertainty studies.

Objective: Develop robust numerical methods for weak-signal analysis, simulation, uncertainty assessment and interpretation of complex datasets.

- **Signal processing:** Developed methods to analyse and separate overlapping low-SNR signals.
- **Simulation:** Built numerical approaches for complex signal environments and large-scale scientific datasets.
- **Diagnostics:** Investigated noise sources and correlations across sensors and simulated observables.
- **Scientific Python:** Developed reproducible Python-based analysis, modelling and validation workflows.
- **Documentation:** Formalised methods, assumptions, results and limitations for collaborative review.

PROJECTS

CATpyLISA – Python platform for model integration, comparison and validation

ESA/NASA LISA space mission

📅 2023 – 2025

Python platform dedicated to heterogeneous model integration, data structuring, comparison of outputs, validation logic and collaborative technical review for a large-scale international scientific mission.

Role: Python development, pipeline design, data structuring, model validation, consistency checks, documentation, technical coordination and contribution consolidation.

Skills: Python, data workflows, model validation, signal processing, statistical modelling, simulation, documentation, GitLab, reproducibility, international collaboration.

AI-assisted engineering, chatbots & agentic workflows

Software productivity / LLM tooling / MCP architectures

📅 2024 – 2026

Use and exploration of AI-assisted development workflows for code review, debugging, documentation, technical synthesis, incremental reasoning, chatbot logic and tool-oriented agent architectures.

Role: prompting, code review support, architecture reasoning, workflow design, documentation and evaluation of AI-assisted productivity patterns.

Skills: LLMs, agents, MCP, chatbot architecture, prompting, Python, software engineering, documentation, productivity tooling.

Noise reduction and predictive modelling workflows

Machine learning / sensor data / precision instrumentation

📅 2021 – 2023

Development of numerical and statistical workflows for weak-signal detection, sensor-data analysis, non-stationary noise reduction and performance comparison across system configurations.

Role: data analysis, statistical modelling, MCMC pipelines, sensor-data interpretation, machine-learning-oriented noise mitigation and performance validation.

Skills: Python, statistics, MCMC, signal processing, ML concepts, deep learning concepts, sensor data, validation, technical reporting.

STRENGTHS

AI architecture, LLMs & acculturation

AI Architecture – ramp-up AI Strategy LLMs Agentic AI MCP Chatbots Prompt Engineering Use-case Framing POC Coordination AI Acculturation Training Workshops Technical Vulgarisation Change Management Best Practices

Data science, machine learning & modelling

Data Science Machine Learning Predictive Analytics Statistical Modelling Bayesian Inference MCMC Regression Classification Model Evaluation Cross-Validation Metrics Feature Engineering Uncertainty Analysis Computer Vision – ramp-up NLP – ramp-up

MLOps / LLMops foundations

MLOps Foundations LLMops – ramp-up CI/CD Versioning Monitoring Reproducible Workflows Model Validation Data Quality Traceability Docker Git / GitLab Linux Security & Compliance – awareness Azure ML – ramp-up Databricks – ramp-up MLflow – ramp-up

Python, software & data engineering

Python NumPy SciPy pandas scikit-learn Matplotlib Jupyter SQL Bash REST API TypeScript Angular Node.js C C++

Systems, validation & industrialisation

Supervised Machine Learning: Regression & Classification

DeepLearning.AI – Andrew Ng

📅 2020

📍 Coursera

Recherche reproductible : principes méthodologiques pour une science transparente

FUN MOOC – Inria

📅 2025

📍 France Université Numérique

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

📅 2024 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, web development, algorithms and software engineering fundamentals.

Complex Systems System Integration Model Integration Validation Strategy IVVQ Anomaly Analysis Root Cause Analysis Technical Reporting Decision Support
Requirements Awareness Risk Analysis Industrial Software Production Support Operational Monitoring

Tools, collaboration & governance

Confluence Jira Zenhub OpenSearch Sentry Power BI Agile Kanban Stakeholder Coordination Documentation Knowledge Sharing International Enviro
Technical English